IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Date of publication xxxx 00, 0000, date of current version xxxx 00, 0000.

Digital Object Identifier 10.1109/ACCESS.2026.0429000

Exploiting Thermal Self-Similarity for Robust Blind Super-Resolution of Real-World Infrared Images

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This research was supported by the Regional Innovation System & Education (RISE) program through the Gyeongbuk RISE Center, funded by the Ministry of Education (MOE) and the Gyeongsangbuk-do, Republic of Korea (2026-RISE-15-115).

ABSTRACT Infrared image super-resolution (IISR) is challenging because real infrared (IR) sensors introduce composite degradations. Optical blur couples with Gaussian noise, fixed-pattern stripe noise, and low-frequency non-uniformity, and their composition and intensity vary with the operating environment and sensor operating time. Existing IISR methods are trained mostly under a single bicubic degradation or handle only a subset of noise types, so they degrade sharply under these diverse real-world conditions. In this paper, we propose a blind super-resolution framework that restores infrared images robustly across the full range of real-world degradations. We first build a physics-based composite degradation model that reproduces the coupling between optical blur and sensor-specific noise, providing a realistic training distribution for blind IISR. We then design a non-local aggregation network (NLAN) that suppresses sensor noise by averaging redundant observations of the same scene content. Thermal scenes are texture-sparse and self-similar, so a given region recurs across the image. NLAN clusters these recurring regions and lets each group exchange information, which cancels the independent sensor noise while preserving the shared structure. This realizes a learnable non-local means that is robust to the composite, out-of-distribution noise of real IR sensors. Finally, a pretrained generative diffusion prior takes the clean restoration and adds back fine textures. Extensive experiments on synthetic benchmarks and real infrared images show that the proposed framework restores images robustly across degradation conditions ranging from weak to severe and produces clean, highly legible restorations of high perceptual quality.

INDEX TERMS Blind super-resolution, clustering attention, composite degradation modeling, diffusion model, infrared image super-resolution, non-local means.

I. INTRODUCTION

INFRARED (IR) imaging captures thermal radiation emitted by objects, enabling scene perception regardless of visible illumination or weather conditions [1]–[3]. This capability has made IR imaging valuable across a wide range of applications, spanning security surveillance, defense systems, autonomous driving, remote sensing, and facility monitoring [4]–[8]. However, IR sensors inherently suffer from lower spatial resolution compared to visible-light sensors due to the physical characteristics of long-wavelength bands, and hardware improvements alone cannot overcome the fundamental trade-off between resolution and sensitivity [9], [10]. To address this limitation, infrared image super-resolution (IISR), which reconstructs high-resolution (HR) images from their low-resolution (LR) counterparts through software-based approaches, has emerged as a key enabling technology.

In the visible-image domain, super-resolution has evolved rapidly from methods assuming idealized bicubic downsampling [11], [12] to sophisticated blind SR frameworks. By employing advanced techniques such as random shuffle pipelines [13] and high-order degradation modeling [14], a mature paradigm for real-world visible SR has been firmly established. However, directly extending this success to the infrared domain is highly non-trivial. Because IR images capture thermal radiation rather than reflected light, they inherently suffer from low contrast, sparse textures, and severely limited high-frequency details [15]. More importantly, real-world IR sensors introduce qualitatively different and highly

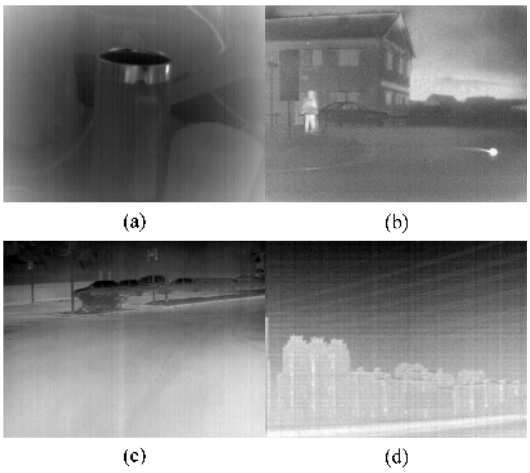


FIGURE 1. Real infrared sensors produce structured, sensor-specific noise such as stripe and Gaussian, unlike the bicubic degradation assumed by most SR methods. (a)–(c) are real degraded thermal images, and (d) is a sample synthesized by our composite degradation model.

complex composite degradations (e.g., optical blur, Gaussian noise, fixed-pattern noise, and non-uniformity noise) that visible-domain models fail to adequately represent [16], [17]. Consequently, blind IISR remains underexplored, lacking a unified framework capable of comprehensively addressing these sensor-specific composite degradations.

Due to the inherent difficulty of modeling these composite degradations, many existing IISR studies have primarily focused on improving network architectures to restore the limited high-frequency details of infrared images, rather than addressing the degradation model itself. In infrared images, the energy in the amplitude spectrum is concentrated at low frequencies, causing severe information overlap between high-frequency details and the low-frequency background [15], [18], which makes it fundamentally difficult to separate and restore fine structural boundaries from noise. To overcome this, frequency-domain-based methods leveraging wavelet transforms [19], [20] and the contourlet transform [18] have been introduced to explicitly exploit spectral representations for high-frequency detail recovery. Transformer-based architectures such as InfraFFN [21] and DASR [22] have also been adopted for IISR, utilizing self-attention mechanisms to capture long-range dependencies and ensure structural consistency across the large, homogeneous thermal regions characteristic of infrared imagery. Furthermore, diffusion model-based restoration approaches [23] have shown promise in recovering high-frequency details. However, the majority of these methods are trained assuming only a single degradation model, namely bicubic downsampling [18]–[20], [23]. As shown in Fig. 1, real infrared sensor environments exhibit composite degradations where blur and various types of noise coexist simultaneously. Consequently, models trained only on bicubic downsampling suffer a severe domain gap, leading to significant performance drops when applied to real-world conditions.

To bridge this gap, blind super-resolution methods addressing real-world degradations have recently begun to emerge in the infrared domain, though they remain limited in scope. For instance, Real-IISR [24] and IRLF-FSRNet [25] tackle specific real-world degradations such as defocus blur or non-uniformity, but they address only isolated degradation factors and fail to comprehensively account for the full composite degradation encountered in real IR sensors. In contrast, DDR-Net [26] models a more comprehensive physics-based composite noise combining Gaussian, stripe, and non-uniformity components. However, as a single-stage regression approach primarily focused on noise removal, it tends to smooth out rather than restore high-frequency textures lost to severe degradation. Consequently, fine details are not adequately recovered, particularly at larger upscaling factors.

Fundamentally, degradation in real-world infrared sensor environments varies dynamically in composition and intensity depending on optical conditions, operating environments, and continuous sensor operation time. In particular, as internal temperatures rise during prolonged operation, sensor noise is progressively exacerbated [16], [17]. A network must therefore stay robust across a wide range of degradation strengths, yet a single network trained to enhance high-frequency detail has no explicit way to separate this noise from real structure. It either amplifies the noise as if it were detail, or suppresses both. Within a single network, robust noise suppression and detail synthesis therefore conflict.

To resolve this, we separate the two goals into a two-stage framework. The first stage, NLAN (Non-Local Aggregation Network), is dedicated solely to recovering a clean image that remains faithful to the scene across degradation strengths. It gathers self-similar regions from across the image and aggregates them: these regions share the same underlying structure but carry independent noise, so aggregating them cancels the noise while preserving the structure. The second stage takes this clean image and uses a pretrained generative diffusion prior to restore the fine textures that heavy degradation had erased. The first stage thus provides robustness and the second detail, without compromising either objective. The main contributions of this work are summarized as follows:

- We adopt and extend a physics-based composite degradation model that synthesizes realistic IR sensor noise, so that the model generalizes across a wide range of real-world degradation intensities.
- We propose NLAN (Non-Local Aggregation Network), which groups self-similar thermal regions from across the entire image and averages the redundant observations within each group, realizing a learnable non-local means that is robust to the composite, out-of-distribution noise of real IR sensors.
- We design a two-stage restoration pipeline that integrates NLAN with a generative diffusion refinement stage, decoupling noise suppression from fine-texture synthesis to reconstruct fine-grained thermal textures at $\times 4$ super-resolution.
- Extensive experiments on both synthetic benchmarks

and real-world infrared images show that our framework attains the best MUSIQ scores in all four evaluation regimes together with competitive MANIQA scores, while its restoration backbone reaches competitive full-reference fidelity, and that it stays robust even when the degradation grows more severe than seen in training.

II. RELATED WORK

A. IMAGE SUPER-RESOLUTION

Since SRCNN [12] pioneered the use of CNNs for single image super-resolution (SISR), the field has undergone a clear architectural evolution. Early works deepened networks through residual learning (VDSR [27], EDSR [28]) and introduced channel attention (RCAN [29]) to selectively amplify informative features. However, these CNN-based approaches are fundamentally constrained by their local receptive fields, limiting the modeling of long-range spatial dependencies. This motivated the adoption of Transformer architectures. SwinIR [11] introduced shifted-window self-attention for efficient global modeling, and subsequent works such as HAT [30] and DAT [31] further expanded the utilized pixel range through hybrid attention and dual spatial–channel aggregation. To curb the quadratic cost of global attention, SPIN [32] and CATANet [33] instead cluster similar tokens by k-means and attend within clusters, approximating non-local attention at a fraction of the cost. While these advances substantially improved reconstruction fidelity, they share a critical assumption that the degradation is known and fixed, typically bicubic downsampling. This assumption creates a fundamental domain gap when confronted with the complex, unknown degradations of real-world images.

B. BLIND SUPER-RESOLUTION

Blind super-resolution (BSR) aims to restore images degraded by unknown processes. A dominant paradigm synthesizes diverse LR–HR pairs through a comprehensive degradation pipeline, training the network to be robust across a wide range of corruptions. BSRGAN [13] first formalized this by randomly shuffling blur, downsampling, and noise operations, and Real-ESRGAN [14] extended it with high-order degradation modeling that cascades the pipeline twice, establishing the dominant framework for practical blind SR. More recently, diffusion priors have advanced BSR: SR3+ [34] combined composite degradation training with noise-conditioning augmentation, ResShift [35] enabled efficient inference via a residual-shifting Markov chain, and DiffBIR [36] decoupled degradation removal from detail regeneration in a two-stage pipeline built on a pretrained latent diffusion model. However, all of these methods are designed for natural RGB images and do not address the fundamentally different degradation characteristics of infrared imagery.

C. INFRARED IMAGE SUPER-RESOLUTION

Infrared images differ fundamentally from visible images in their spectral characteristics, with low contrast, homogeneous backgrounds, and severely attenuated high-frequency content due to the physics of thermal radiation [15]. These properties make the direct application of visible-domain SR methods suboptimal and have motivated IR-tailored approaches, particularly frequency-domain methods that explicitly address the spectral imbalance of IR data. IRSRMamba [20] and GPS-Mamba [19] integrate wavelet and spectral representations within state-space models, while CoRPLE [18] leverages the contourlet transform for multi-directional high-frequency extraction. Transformer-based architectures (InfraFFN [21], DASR [22]) combine convolution and self-attention to jointly model local textures and global thermal structure, and DifISR [23] introduces diffusion-based restoration with gradient guidance tailored to IR characteristics. However, nearly all of these methods assume bicubic downsampling as the sole degradation [18]–[20], [23]. A few recent blind IISR methods have begun to move toward real degradations. RealIISR [24] constructs a real-captured LR–HR dataset and progressively removes optical and motion blur through an autoregressive restoration guided by thermal structure. IRLFFSRNet [25] decomposes structure and non-uniform noise into distinct frequency components and learns their cross-frequency correlations, performing super-resolution and non-uniformity correction jointly within a single model. DDRNet [26] incorporates a physics-based composite noise model spanning Gaussian, stripe, and non-uniformity components into training-time synthesis, and unifies denoising and super-resolution through a dual-stream architecture that processes low- and high-frequency components separately. While these methods mark clear progress by explicitly accounting for IR-specific real degradations, each covers a differently restricted portion of the degradation space.

D. INFRARED IMAGE ENHANCEMENT

Independently of SR, a substantial body of work addresses IR image quality enhancement at the original resolution. The noise of IR sensors is distinct from that of visible cameras and can be decomposed into several physically grounded components. These include Gaussian noise from thermal agitation in the focal-plane array and readout electronics, fixed-pattern noise (FPN) manifesting as stripe artifacts from non-uniform detector response, and low-frequency non-uniformity noise from thermal self-emission of internal optics [16], [17]. Deep learning methods have addressed these components along separate lines. To remove broadband and fixed-pattern noise, [37] and [16] employ residual attention CNNs and physics-inspired noise modeling, respectively. To suppress directional stripes and correct real non-uniformity, [38] introduces a stripe-aware attention mechanism that recovers the stripe directionality disrupted by Gaussian noise. Beyond noise removal, [39] and [40] further improve contrast and visibility through dynamic-range compression and physics-guided enhancement. However, these methods operate exclusively at the native sensor resolution and do not address the simultaneously coexisting low-resolution problem. Our work unifies both perspectives by jointly modeling optical blur and sensor-specific composite noise within a single blind SR framework.

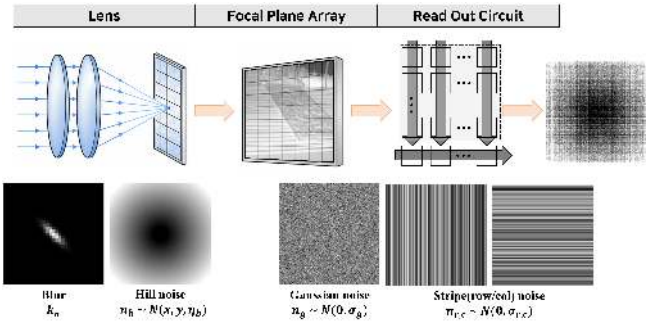


FIGURE 2. Our physics-based composite degradation model, organized along the infrared imaging chain. Each corruption is injected at the stage where it physically arises: optical blur and low-frequency hill bias at the lens, Gaussian read noise at the focal-plane array, and row/column stripe (fixed-pattern) noise at the readout circuit, producing the degraded image on the right.

III. REALISTIC INFRARED DEGRADATION MODELING

To train an SR model that generalizes effectively to real-world degraded infrared images, we synthesize LR–HR training pairs using a pipeline that strictly follows the physical imaging chain of actual IR sensors. Standard blind SR pipelines [13], [14], [41] combine blur, downsampling, and generic noise (e.g., Gaussian and JPEG artifacts) designed for natural RGB images. While effective for visible-light images, this approach fails to capture the unique noise characteristics—particularly stripe artifacts and low-frequency non-uniformity—that dominate actual thermal imagery. Instead, we inject each specific corruption at the exact stage it occurs along the IR imaging pathway (i.e., optics, focal plane sampling, and readout). The degraded LR image y is generated from the clean HR image x as follows:

$$y = \left(x \otimes k_\sigma + n_h\right)\downarrow_s + n_r + n_c + n_g, \tag{1}$$

where k_σ denotes a Gaussian blur kernel, $\downarrow_s$ represents downsampling by a scale factor of s, n_h is the low-frequency hill bias, n_r and n_c indicate row and column stripe noise respectively, and n_g is the Gaussian noise. The grouping in this formulation mirrors the physical imaging chain: the optical terms act on the continuous pre-sampling scene, while the detector and readout terms are introduced in the discrete low-resolution domain, as detailed below. The sensor noise is injected after the downsampling because the readout operates on the native detector grid, whose output is the low-resolution image itself, and injecting it before the downsampling would low-pass and spatially correlate the noise, contradicting the per-pixel statistics measured from real captures in Section VI-A. Fig. 2 visualizes each component alongside the synthesized composite result.

Optical stage. Thermal radiation initially passes through the lens, where imperfect focal alignment blurs the scene. We model this phenomenon using a Gaussian defocus kernel k_σ. Moreover, the optical system itself contributes radiation. Due to the inherent non-uniformity of optical elements and variations in the ambient temperature field, stray thermal self-emission from the lens barrel and housing reaches the detector [42], [43]. This introduces a hill bias n_h—a low-frequency brightness variation that shifts smoothly across the image, typically manifesting as a dark center with brighter edges. This bias accumulates during prolonged continuous operation and persists when on-chip non-uniformity correction is inadequate. Following heat transfer-based formulations for ambient sensor temperature fields [16], [42], [43], we model n_h as a radial bias determined by the spatial distance of each pixel from the sensor's center. Because both k_σ and n_h occur prior to spatial sampling, they are applied directly to the HR scene before the signal is downsampled via $\downarrow_s$.

Focal plane and readout stage. After the radiation reaches the infrared focal plane array (IRFPA), the detectors and their readout electronics add two further types of noise. First, the thermal motion of electrons in the readout circuitry adds a small random noise. Because it comes from the circuitry rather than the scene, this readout noise is independent of the signal and spread evenly across the image; we model it as Gaussian noise n_g [16], [44]. Second, the readout circuit (ROIC) adds a structured, fixed-pattern error. Its column-parallel amplifiers are never perfectly matched, so each column carries a slightly different offset that appears as vertical stripes (n_c); likewise, timing and supply-voltage fluctuations in the row-by-row scanning add a per-row offset seen as horizontal stripes (n_r) [17], [45]. Because each offset stays constant along its stripe direction, the resulting pattern is highly structured and strongly directional [46].

Composite and random degradation. In real uncooled thermal imaging these corruptions generally co-exist, but their relative dominance and intensity vary with the operating conditions: hill bias grows as the sensor heats up over prolonged operation, while stripe and Gaussian noise shift with the ambient temperature. Rather than assuming a single fixed degradation profile, we randomly vary both the presence and the intensity of each component for every training image, exposing the network to the full spectrum from near-clean to severely degraded inputs. This prevents overfitting to any specific degradation pattern and encourages robust generalization. Detailed parameter ranges are provided in the implementation details. The physical fidelity of this composite model is validated against measurements from a real infrared sensor in Section VI-A.

IV. PROPOSED METHOD

As motivated in Section I, robust noise suppression and fine-texture synthesis conflict within a single network, so we decouple them into a two-stage framework, *NLAN-Diff*. The first stage, *NLAN* (Non-Local Aggregation Network), groups self-similar regions across the image and aggregates them, so that the independent sensor noise averages out while the shared content remains, yielding a clean restoration that is faithful across degradation strengths. The second stage conditions a pretrained generative diffusion prior on this clean image to synthesize the fine texture that heavy degradation had erased. To support this division of labor, we relax the lightweight design of earlier clustering attention and give the network

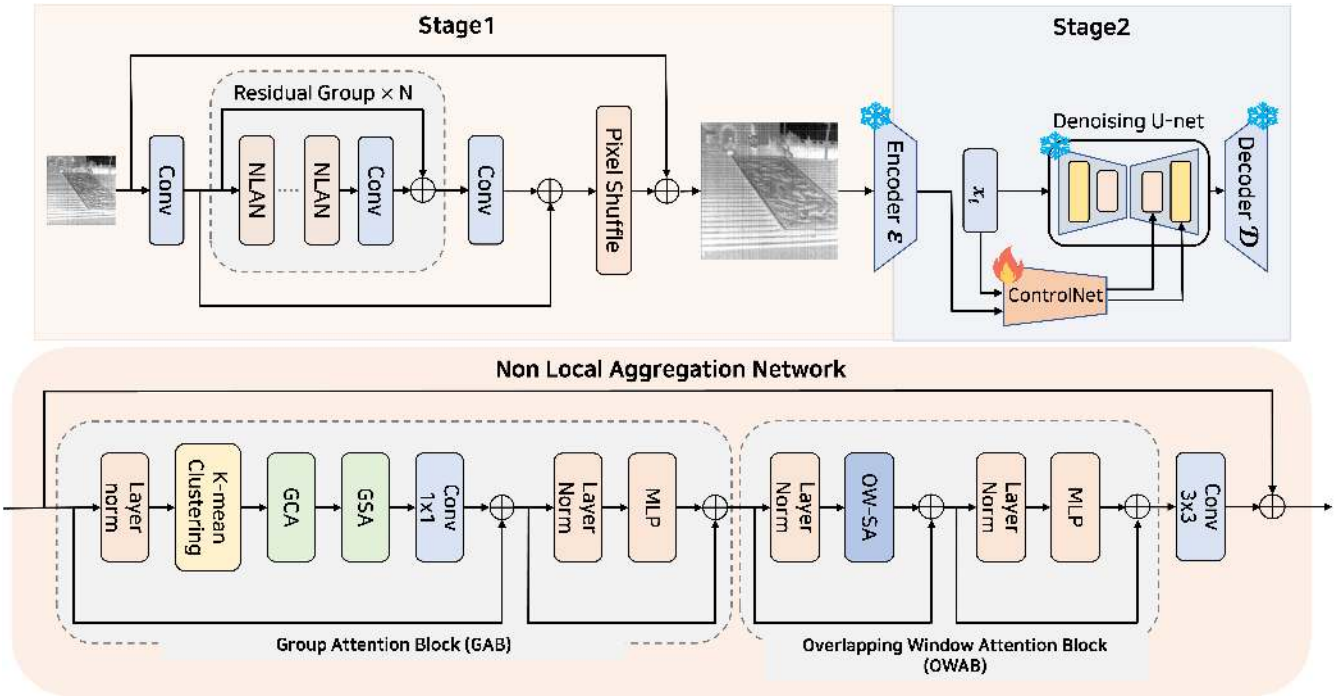


FIGURE 3. **Overall architecture of the proposed two-stage NLAN-Diff framework. Stage 1, NLAN (Non-Local Aggregation Network), is a clustering-based restoration network that produces a clean high-resolution image. A shallow convolution feeds *N* Residual Groups, each pairing a Group Attention Block (GAB) with an Overlapping Window Attention Block (OWAB), followed by a PixelShuffle reconstruction with a global residual over the bilinearly upsampled input. The GAB clusters tokens by similarity, then lets each token attend to a set of global centers (GCA) and to the other tokens of its group (GSA). The OWAB applies overlapping window self-attention. Stage 2 refines the result with a frozen diffusion prior controlled by a trainable ControlNet.**

more restoration capacity.

A. OVERVIEW

We now describe Stage 1, which produces the clean restoration.

Given an input $I_{LR} \in \mathbb{R}^{H\times W\times C_{in}}$, a single 3×3 convolution embeds it into a feature map of width C,

$$F_0 = \mathrm{Conv}_{3\times 3}(I_{LR}) \in \mathbb{R}^{H\times W\times C}. \tag{2}$$

The body applies N Residual Groups (RGs). Each RG refines a token first *non-locally* and then *locally*. The Group Attention Block (GAB) groups similar regions from anywhere in the image and lets them exchange information, and the Overlapping Window Attention Block (OWAB) then sharpens the immediate neighborhood. Writing F for the RG input,

$$F' = F + \mathrm{Conv}_{3\times 3}\big(\mathrm{OWAB}(\mathrm{GAB}(F))\big), \tag{3}$$

where the GAB clusters tokens and exchanges information across and within groups, and the OWAB applies overlapping window self-attention. After N RGs, a long skip adds F_0 back to the deep feature F_N, and the reconstruction H_{rec} (convolutions with a PixelShuffle upsampler [47]) predicts a residual over the bilinearly upsampled input:

$$I_{SR}^{(1)} = \mathrm{Bilinear}_{\times s}(I_{LR}) + H_{rec}(F_0 + F_N). \tag{4}$$

As in standard SR networks, predicting this residual rather than the full image eases optimization and lets the blocks focus their capacity on high-frequency detail. $I_{SR}^{(1)}$ is the Stage-1 clean restoration, refined into the final detailed result I_{SR} by Stage 2.

B. GROUP ATTENTION BLOCK

Suppressing signal-independent sensor noise without hallucinating detail is fundamentally a problem of averaging multiple noisy observations of the same signal, where the uncorrelated noise cancels while the shared signal is preserved. This is the principle behind classical non-local denoisers such as non-local means (NLM) [48] and BM3D [49]. Thermal imagery is especially amenable to it, since thermal scenes are texture-sparse and piecewise-uniform, so a region such as sky or road recurs across the image and provides abundant redundant observations of the same content. Inspired by this principle, we design the non-local block as a learnable form of non-local means, in which similar tokens drawn from the entire image are attended together so that the independent noise averages out while the shared content is preserved. Unlike classical NLM, which weights neighbors by fixed patch distances, both the neighborhood and the weights are learned here, so they adapt to the unknown degradation encountered at test time.

To form such groups, we cluster the tokens by content similarity with a k-means, so that tokens belonging to the same recurring region are pooled together regardless of their position. For blind SR, where the degradation is unknown at test time, this clustering-based non-local averaging is attractive because it removes noise without assuming a fixed noise model. The

GAB applies k-means clustering, then group cross-attention (GCA), then group self-attention (GSA), described below.

K-means clustering. The clustering is what lets the network exploit self-similarity. Since tokens are grouped by content similarity, the recurring instances of a region, however far apart in the image, are placed in the same group and can then be averaged. Following CATANet [33], we group the tokens by a cosine k-means. Unlike that work, where the grouping approximates global attention at a lower cost, here it selects the redundant observations that the within-group averaging then combines. Let $\{f_i\}_{i=1}^{HW}$ be the tokens and $\{c_j\}_{j=1}^{M}$ a set of M centers, initialized by average pooling over regular regions. Each token is assigned to its most similar center,

$$g(i) = \arg\max_j \; s(f_i, c_j), \qquad \mathcal{G}_j = \{\, i \mid g(i) = j \,\}, \tag{5}$$

where $s(\cdot,\cdot)$ is the cosine similarity and $\mathcal{G}_j$ is the set of tokens assigned to center j. During training, the centers are refined within each image by a few k-means update steps, and accumulated across images into a *single, global* set by an exponential moving average [33], [50]. At inference, this global set is frozen. Accumulated in this way, the centers represent thermal structures that recur across the training set rather than the content of any single frame, and they remain stable when the test input is heavily degraded. Freezing does not remove the input dependence of the averaging, since the assignment in Eq. (5) and the within-group attention are recomputed for every test image.

The clustered tokens are then refined in two ordered steps that play complementary roles. GCA brings in global, cluster-level context, and GSA then denoises by averaging within each group.

Group cross-attention (GCA). *First*, every token cross-attends to the M cluster centers, using the token as the query and the centers as the keys and values. Each center is the average of the tokens in its cluster, so it summarizes the content of that cluster, such as a smooth background or a uniform thermal region. Through the softmax, a token attends most strongly to the center whose content best matches its own, so a background token, for instance, is drawn toward the background prototype. Each token thereby gains a global, cluster-level context, before the within-group attention of the next step averages the matching tokens of the same image to cancel their independent noise. Stacking the tokens as $X = [f_1, \ldots, f_{HW}]^\top$ and the centers as $C = [c_1, \ldots, c_M]^\top$, we compute

$$Q = XW_q^c, \qquad K_c = CW_k^c, \qquad V_c = CW_v^c, \tag{6}$$

and update each token by attending to the centers,

$$\tilde{X} = X + \mathrm{softmax}\!\left(\frac{QK_c^\top}{\sqrt{d}}\right)V_c, \tag{7}$$

where d is the head dimension.

Group self-attention (GSA). *Then*, the tokens attend within their own group. A group collects similar tokens from across the whole image, which under the self-similarity of thermal scenes are repeated, independently noisy observations of the same underlying content. Averaging them follows the non-local means principle: the shared content reinforces while the independent noise cancels. For a group $\mathcal{G}$, we stack the GCA-output tokens assigned to it as $\tilde{X}_\mathcal{G}$. With

$$Q_\mathcal{G} = \tilde{X}_\mathcal{G}W_q, \qquad K_\mathcal{G} = \tilde{X}_\mathcal{G}W_k, \qquad V_\mathcal{G} = \tilde{X}_\mathcal{G}W_v, \tag{8}$$

the within-group attention takes their weighted average,

$$\hat{X}_\mathcal{G} = \mathrm{softmax}\!\left(\frac{Q_\mathcal{G}K_\mathcal{G}^\top}{\sqrt{d}}\right)V_\mathcal{G}. \tag{9}$$

Eq. (9) scales quadratically with the group size, and in the worst case a single cluster absorbs most of the tokens, for instance when a large homogeneous thermal region dominates the image, so that a single group can hold almost all N tokens and the cost approaches $O(N^2)$. We therefore sort the tokens by cluster index and partition them into fixed-size groups of g_s tokens, which caps the attention span at g_s and gives an $O(Ng_s)$ cost that is linear in N and independent of the cluster-size distribution.

The denoised tokens are reassembled over all groups and added back to the block input through the residual below, so the GSA output itself carries no separate residual. Finally, a 1×1 convolution mixes the aggregated channels and a feed-forward network refines the result, each forming a residual sub-layer of the block as in the OWAB below. With F the GAB input and $\hat{X}$ the GSA output reassembled over all groups,

$$\hat{F} = F + \mathrm{Conv}_{1\times1}(\hat{X}), \tag{10}$$

$$\mathrm{GAB}(F) = \hat{F} + \mathrm{MLP}(\mathrm{LN}(\hat{F})). \tag{11}$$

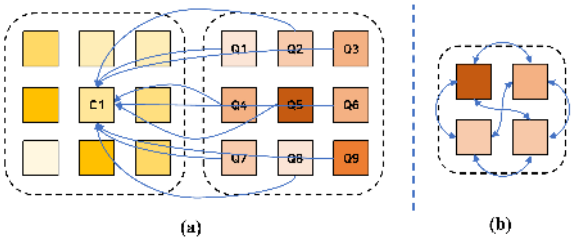


FIGURE 4. The two attentions inside the Group Attention Block (GAB). (a) Group cross-attention (GCA), where every token (Q1–Q9) attends to the global cluster centers (e.g., C1), pulling each token toward its matching center. (b) Group self-attention (GSA), where tokens within the same group attend to one another, averaging out independent noise while preserving the shared content.

C. OVERLAPPING WINDOW ATTENTION BLOCK

The GAB removes noise by aggregating non-local information, and the OWAB then restores the fine local detail within each neighborhood. Prior clustering-based SR keeps this local module deliberately lightweight to save computation [32], [33]. Since we target robust restoration of real IR images rather than computational efficiency, we adopt a stronger local block based on overlapping window attention [30]. Standard window self-attention [11] splits the feature into non-overlapping windows and attends within each, so a window

cannot see beyond its own border. The overlapping variant keeps the query inside its window but draws the key and value from a larger region that spills into the neighboring windows, so each query attends across window boundaries and the effective receptive field grows. Concretely, for a window with tokens X_w and its surrounding overlapping region with tokens X_o, we compute

$$Q = X_w W_q^o, \quad K = X_o W_k^o, \quad V = X_o W_v^o, \tag{12}$$

and the overlapping-window attention is

$$\text{OW-MSA} = \text{softmax}\Big(\frac{QK^\top}{\sqrt{d}} + B\Big)V, \tag{13}$$

where B is a relative position bias. Applied over every window, this attention and a feed-forward network form two pre-norm residual sub-layers,

$$\hat{F} = F + \text{OW-MSA}(\text{LN}(F)), \tag{14}$$

$$\text{OWAB}(F) = \hat{F} + \text{MLP}(\text{LN}(\hat{F})). \tag{15}$$

The OWAB thus sharpens local detail while the GAB suppresses non-local noise, so the two passes of each Residual Group are complementary.

D. STAGE 2: GENERATIVE DIFFUSION REFINEMENT

The first-stage network restores a clean HR image with suppressed noise, but, being trained with a pixel-based loss, may lack realistic fine textures. The second stage employs a pretrained latent diffusion model as a generative prior to recover such textures. Following DiffBIR [36] and StableSR [51], we use a pretrained Stable Diffusion [52] as the backbone and inject the first-stage output $I_{SR}^{(1)}$ as a condition through a ControlNet-based control branch [53]. This stage depends critically on its conditioning input. When the conditioning image still carries sensor noise, the diffusion prior mistakes that noise for real structure and sharpens it into false detail, so it must be given a clean condition to begin with. Rather than a generic restoration backbone, we condition on NLAN, which is designed to suppress the composite, out-of-distribution noise of real IR sensors and thus supplies a clean, structurally faithful reconstruction. This decoupling lets the generative prior focus solely on synthesizing realistic fine textures rather than simultaneously removing sensor-specific degradations. The VAE encoder/decoder and the U-Net denoiser of Stable Diffusion are frozen, and only the ControlNet is trained with the standard latent-space denoising objective,

$$\mathcal{L}_{\text{Diff}} = \mathbb{E}_{z_0,t,\epsilon}\big[\|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2\big], \tag{16}$$

where $z_0 = \mathcal{E}(I_{HR})$ is the VAE latent of the ground-truth image, $z_t = \sqrt{\bar{\alpha}_t}\, z_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon$ ($\epsilon \sim \mathcal{N}(0, I)$) the noised latent at step t, ϵ_θ the ControlNet-controlled denoiser, and c the condition derived from the first-stage output $I_{SR}^{(1)}$ (and a text prompt). At inference, the reverse diffusion process adds realistic textures to the first-stage output, yielding the final result I_{SR}.

V. EXPERIMENTS

A. EXPERIMENTAL SETTINGS

Datasets. We train on the HR images of the FLIR-IISR dataset released by [24] (1,457 images) and evaluate across three degradation regimes, from controlled-synthetic to real in-the-wild conditions. **(i) Synthetic.** The HR images of FLIR (130) and M^{3}FD [55] (20) are corrupted by our composite degradation pipeline (Section III). Since the pipeline randomizes each corruption per image, this regime spans a wide range of difficulty from easy to severe. **(ii) Real-Light.** This set contains 47 real infrared images from IR100 [56] and SIRST-v2 [57] that are lightly degraded, with some near-clean samples. **(iii) Real-Heavy.** This set contains 29 more heavily degraded real infrared images, gathered from our self-captured frames, classic fusion benchmarks [58], [59], and the FLIR-ADAS dataset [60]. Because most of these sources are video sequences whose consecutive frames are nearly identical, we manually select one frame per distinct scene, so the two sets consist of mutually non-redundant scenes rather than hundreds of correlated frames. The two real regimes have no ground truth and are evaluated with no-reference metrics.

Degradation Synthesis. Rather than relying on a fixed bicubic kernel, we synthesize LR–HR training pairs on the fly following the physics-based composite model of Section III, in which each corruption is injected at its originating stage of the imaging chain, and each noise component is applied independently with its own probability, so that arbitrary subsets co-occur with varying intensity. Concretely, for every HR patch we apply five operations. *Optical blur* is always active, drawn from a five-type Gaussian kernel mixture (isotropic, anisotropic, generalized isotropic/anisotropic, and plateau) with kernel size sampled uniformly from $\{7, 9, \ldots, 21\}$ and width $\sigma \in [0.2, 1.5]$. A *low-frequency hill bias* (probability 0.4) is modeled as a radial bowl with amplitude $\eta \in [2, 25]/255$. The $\times 4$ *downsampling* draws its resampling operator at random among bicubic, bilinear, and area. *Stripe noise* is added in the LR domain as independent per-row (probability 0.3, $\sigma_r \in [1, 3]/255$) and per-column (probability 0.4, $\sigma_c \in [1, 3]/255$) Gaussian offsets. *Gaussian read noise* (probability 0.8) is i.i.d. with $\sigma \in [0, 5]/255$, which leaves a fraction of the training pairs near-clean. The optical terms, blur and hill bias, are applied to the HR scene before downsampling, whereas the sensor terms, stripe and read noise, are injected afterwards, faithfully reproducing the order in which real IR sensors introduce each degradation. This wide, randomized degradation space exposes the network to the full spectrum of mild-to-severe real-world conditions during training.

Implementation Details. We focus on $\times 4$ super-resolution throughout this work. NLAN uses a feature width of $C = 120$ and $N = 8$ Residual Groups, each pairing a GAB with an OWAB. The GAB uses 6 attention heads, with (M, g_s) set to $(16, 256)$, $(32, 128)$, $(64, 64)$ and $(128, 32)$ over successive Residual Groups and this schedule repeated once across the eight groups, and the OWAB uses a window size of 16 with an overlap ratio of 0.5. The grayscale input is replicated to three

TABLE 1. Overall comparison: no-reference perceptual quality (MUSIQ↑, MANIQA↑, NIQE↓) for the final ×4 outputs across four regimes. Synthetic (FLIR, M³FD) uses our composite degradation, while Real-Light and Real-Heavy are real images without ground truth. **Best** and second best.

	Synthetic degradation						Real-world degradation					
	FLIR			M³FD			Real-Light			Real-Heavy		
Method	MUSIQ↑	MANIQA↑	NIQE↓	MUSIQ↑	MANIQA↑	NIQE↓	MUSIQ↑	MANIQA↑	NIQE↓	MUSIQ↑	MANIQA↑	NIQE↓
RRDBNet [54]	41.30	0.251	8.13	35.01	0.248	8.19	29.93	**0.208**	9.80	27.87	0.219	7.91
SwinIR [11]	41.35	0.248	7.86	35.06	0.244	7.90	30.82	0.207	9.55	28.53	0.222	7.78
DAT [31]	37.60	0.235	8.17	32.21	0.234	8.26	28.87	0.203	9.80	27.47	0.221	7.89
HAT [30]	40.08	0.247	8.23	34.90	0.246	8.19	30.57	0.204	9.54	28.24	0.225	7.84
CATANet [33]	37.93	0.232	8.36	33.01	0.232	8.25	29.41	0.202	9.58	27.69	0.222	7.96
IRLF-FSRNet [25]	39.76	0.244	8.39	34.65	0.243	8.29	29.75	**0.208**	9.74	28.47	0.223	7.83
DDR-Net [26]	34.62	0.227	8.06	29.90	0.229	8.06	27.32	0.201	9.28	26.62	**0.227**	7.73
BSRGAN [13]	43.68	0.235	6.95	36.37	0.225	6.83	30.70	0.203	8.70	31.29	0.208	6.12
Real-ESRGAN [14]	45.37	0.256	6.88	37.38	0.239	7.07	31.58	0.201	8.36	30.86	0.204	6.45
ResShift [35]	41.59	0.226	7.26	35.60	0.222	6.95	31.94	0.199	8.41	30.54	0.214	7.02
NLAN-Diff (Ours)	**47.20**	**0.276**	**6.28**	**42.85**	**0.280**	**6.02**	**33.13**	0.204	**8.12**	**33.69**	0.224	**5.42**

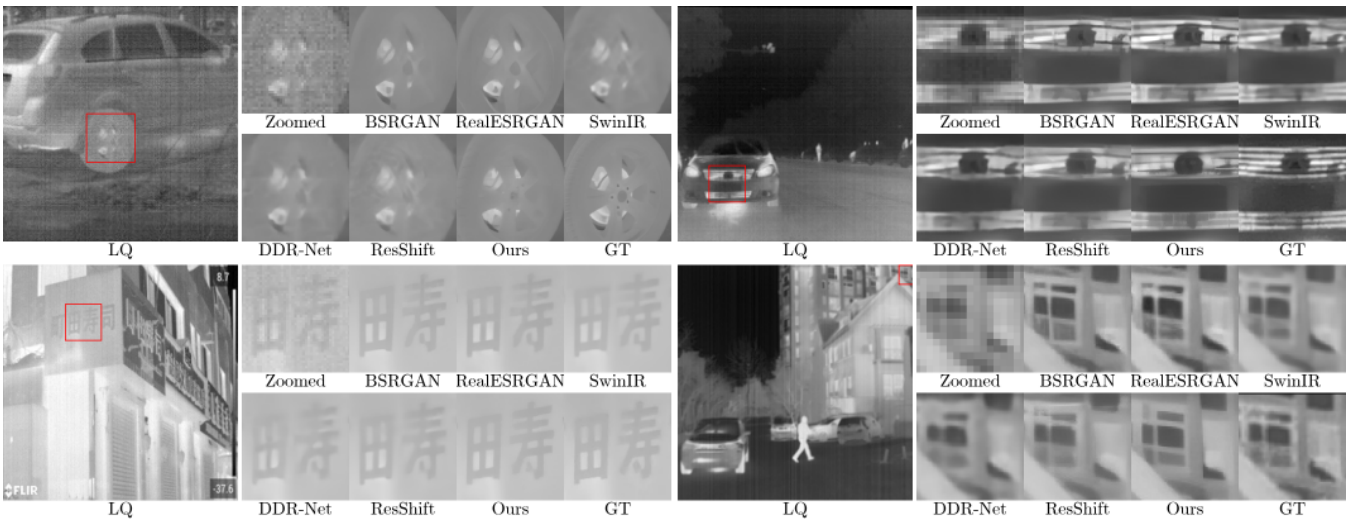


FIGURE 5. Qualitative comparison on synthetic infrared images (FLIR, M³FD).

channels so that the model shares the same input/width configuration as the SwinIR/HAT baselines for a fair comparison. We train Stage 1 on 256×256 HR patches (yielding 64×64 LR inputs at $\times 4$) with random flipping and rotation, using the Adam optimizer [61] ($\beta_1 = 0.9$, $\beta_2 = 0.99$) with an initial learning rate of 2×10^{-4} halved at $\{100k, 160k, 180k, 190k\}$ iterations via a multi-step schedule, a batch size of 12, for a total of 200,000 iterations, and optimized with an ℓ_1 pixel loss. For the second stage, we adopt Stable Diffusion 2.1-base as the backbone, freeze its VAE and U-Net, and train only the ControlNet branch on the first-stage outputs with the objective of Eq. (16), using AdamW with a learning rate of 1×10^{-5} and a batch size of 4 for 200,000 iterations. At inference, we run 50 denoising steps with an empty text prompt. All models are trained in PyTorch on a single NVIDIA RTX 4090 GPU.

Evaluation Metrics. We adopt two complementary protocols. For the overall comparison of final outputs, where the real sets have no ground truth, we use no-reference metrics that predict perceptual quality directly from a single image without any reference: MUSIQ [62], a multi-scale transformer-based quality model, MANIQA [63], an attention-based one (higher is better for both), and the statistics-based NIQE [64], for which a lower score means better quality. For the backbone comparison on the synthetic sets, where ground-truth HR is available, we additionally report full-reference metrics: PSNR and SSIM, which measure pixel- and structure-level fidelity against the ground truth, and LPIPS [65], which measures perceptual distance in a deep feature space (lower is better).

Comparison Methods. We compare against two families of methods, all trained on the FLIR training set under the identical composite degradation for a fair comparison. The first family is *restoration* (regression) SR networks that output a single deterministic reconstruction, namely the CNN-based RRDBNet [54], the transformer-based SwinIR [11], DAT [31], and HAT [30], the clustering-based CATANet [33], and the infrared-specific IRLF-FSRNet [25] and DDR-Net [26]. The second family is *generative/blind* SR pipelines that target real-world degradation, namely the GAN-based BSRGAN [13] and Real-ESRGAN [14], and the diffusion-based ResShift [35]. We report two complementary comparisons. The *final-output* comparison (Table 1) evaluates the final HR results of all methods with no-reference metrics across the synthetic and real regimes. The *backbone* comparison

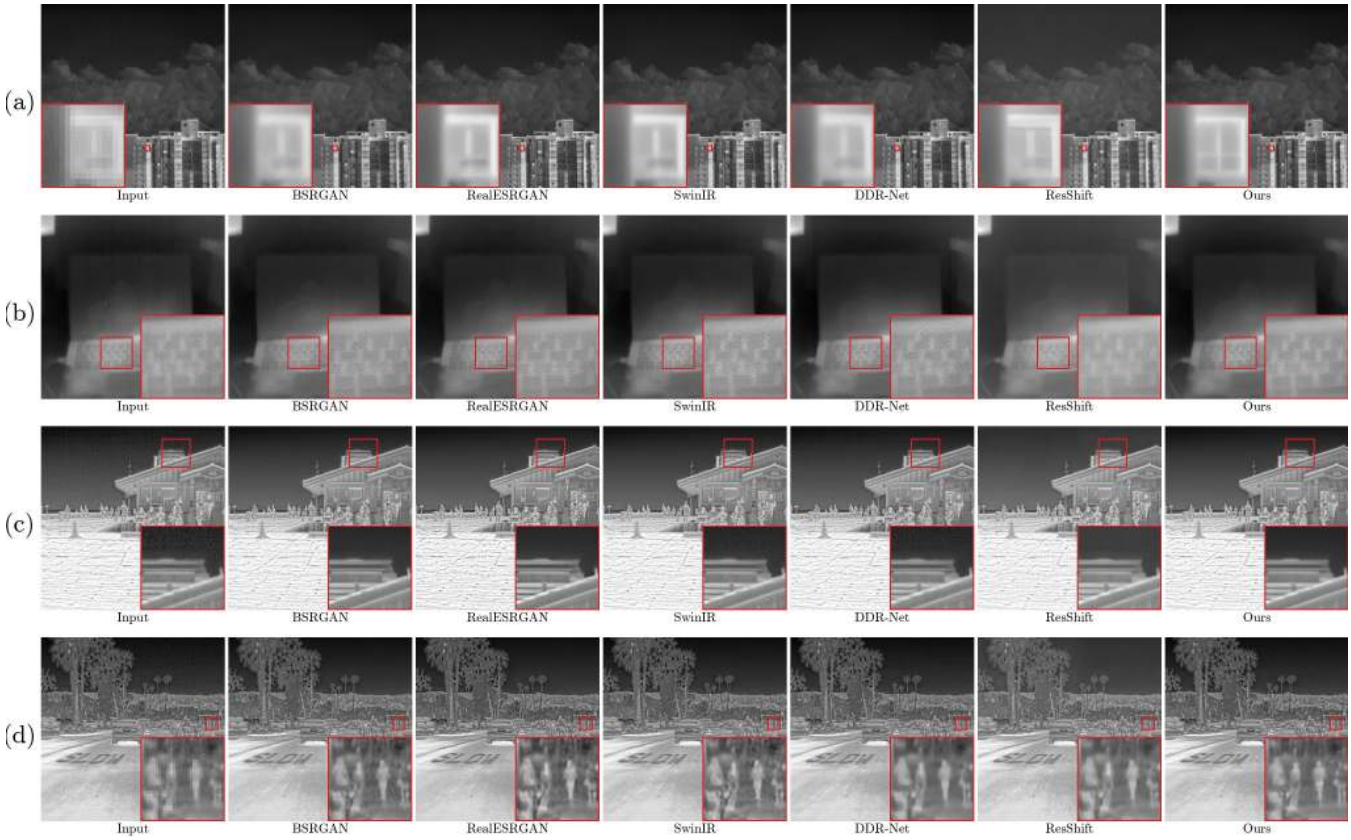


FIGURE 6. **Qualitative comparison on real infrared images (no ground truth). The rows span different real degradation conditions: (a) an almost noise-free input, (b) real degradation with both Gaussian and stripe noise, and (c)–(d) real degradation dominated by strong Gaussian noise. NLAN-Diff stays clean and detailed across all of them, while the other methods leave residual stripes and grain.**

(Table 2) instead evaluates the restoration networks, together with our Stage-1 NLAN, at full image resolution against the ground truth on the synthetic sets, isolating the quality of the clean reconstruction that conditions the diffusion stage.

B. OVERALL COMPARISON

Quantitative. Table 1 reports the no-reference perceptual quality (MUSIQ, MANIQA, NIQE) of the final outputs across the four regimes, from synthetic (FLIR, M^3FD) to real in-the-wild (Real-Light, Real-Heavy). Because our final stage is generative and its output is not pixel-aligned to the ground truth, we use no-reference metrics so that regression and generative methods are compared on the same footing. NLAN-Diff achieves the best MUSIQ and the best NIQE in all four regimes and the best MANIQA on both synthetic sets. Its synthetic MUSIQ reaches 47.20 on FLIR and 42.85 on M^3FD, clearly above the second-best method (Real-ESRGAN, 45.37 and 37.38). On the two real regimes it again leads MUSIQ—33.13 on Real-Light and 33.69 on Real-Heavy, ahead of the second best by 1.19 and 2.40—while staying competitive on MANIQA. Overall, NLAN-Diff restores robustly and consistently across both synthetic and real images and from mild to severe sensor noise, attaining the highest perceptual quality where the baselines degrade as the degradation grows.

Qualitative. Figs. 5 and 6 show visual comparisons on the synthetic and real sets. On the synthetic images (Fig. 5), where the ground truth is available for reference, NLAN-Diff recovers sharp edges and fine thermal texture that most closely matches the ground truth, while the other methods recover less detail. The advantage becomes more evident on the real images (Fig. 6), whose degradation is unknown and out-of-distribution. We show four representative conditions, from an almost noise-free input (a) to inputs with combined or strong sensor noise (b)–(d): under these unseen degradations the baselines tend to leave residual noise or lose fine detail, whereas NLAN-Diff stays clean and detailed across all four conditions. By removing the stripe and grain noise that obscures the other reconstructions, it yields the most legible thermal images, keeping edges and small thermal targets clearly visible where the competing methods leave them buried in noise. Although the diffusion prior is pretrained on visible-light images, the textures it synthesizes across Figs. 5 and 6 remain consistent with thermal signatures, and we observe no visible-domain patterns such as color-texture artifacts, which we attribute to conditioning the prior on the noise-suppressed NLAN restoration with an empty text prompt. This consistency is the main practical advantage of our two-stage design on real-world infrared images.

TABLE 2. Backbone comparison: full-reference fidelity (PSNR ↑, SSIM ↑, LPIPS ↓) on the synthetic sets, isolating the quality of the first-stage restoration that conditions the diffusion stage. Best and second best. Real-world behavior is shown qualitatively in Fig. 7.

Method	FLIR PSNR ↑	FLIR SSIM ↑	FLIR LPIPS ↓	M^3FD PSNR ↑	M^3FD SSIM ↑	M^3FD LPIPS ↓
RRDBNet [54]	35.92	0.956	0.1397	36.43	0.959	0.1828
SwinIR [11]	**36.50**	**0.958**	**0.1304**	36.97	0.959	0.1716
DAT [31]	35.62	0.953	0.1442	37.27	**0.960**	**0.1694**
HAT [30]	35.71	0.954	0.1379	37.16	**0.960**	0.1732
CATANet [33]	35.20	0.950	0.1482	37.07	0.958	0.1754
IRLF-FSRNet [25]	35.70	0.952	0.1468	37.16	**0.960**	0.1770
DDR-Net [26]	34.58	0.944	0.1729	36.50	0.953	0.1931
Ours (Stage-1)	35.96	0.955	0.1338	**37.29**	**0.960**	**0.1694**

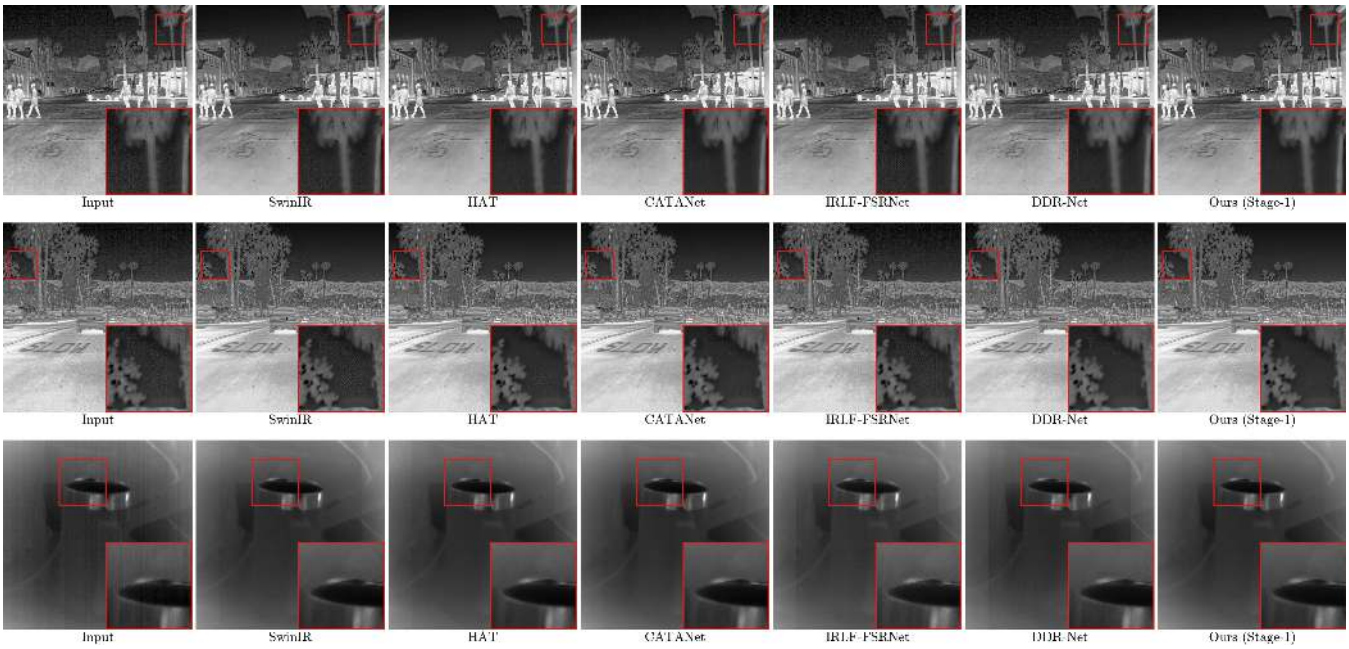


FIGURE 7. Backbone comparison on heavily degraded real infrared images (Real-Heavy). Each panel is a full view with a zoom crop at the corner; NLAN (Ours, Stage-1) suppresses the real sensor noise that the other backbones leave behind or amplify.

C. BACKBONE COMPARISON

Quantitative. To isolate the contribution of the first stage, Table 2 compares our Stage-1 network (NLAN) against representative restoration backbones at full image resolution on the synthetic sets, where the ground truth is available and we report full-reference fidelity (PSNR, SSIM, LPIPS). The aim here is not to win every metric but to verify that NLAN yields a clean, faithful restoration suitable to condition the diffusion stage. NLAN ranks among the top backbones: it attains the best PSNR on M^3FD and the second-best PSNR and LPIPS on FLIR, behind only SwinIR, with all the leading methods within a narrow margin. This shows that on in-distribution synthetic degradations NLAN reconstructs the scene as faithfully as the strongest backbones, and it does so with far fewer parameters—about 4.7M, versus 11.9M for SwinIR and 20.8M for HAT. The decisive difference instead appears on real inputs, whose degradation is out-of-distribution and has no ground truth to measure against; we therefore examine the real-world behavior qualitatively in Fig. 7.

Qualitative. Fig. 7 compares the restoration backbones on heavily degraded real images (Real-Heavy), whose noise is not seen during training. Tuned to the synthetic degradation, the other backbones leave residual sensor noise or amplify it on these out-of-distribution inputs, whereas NLAN suppresses the stripe and grain and keeps the structure clean. This robustness follows from its global clustering: noisy test tokens are pulled toward the dataset-level clean prototypes instead of being processed in isolation. The clean restoration is precisely what the diffusion stage needs as a reliable condition, which is why our two-stage design holds up on real infrared images where single-stage backbones do not.

D. ABLATION STUDY

Effect of GAB and OWAB. In every Residual Group the Stage-1 network pairs two attention blocks that differ in *how they aggregate tokens*. The Overlapping Window Attention Block (OWAB) applies self-attention within a small local spatial window, whereas the Group Attention Block (GAB) first clusters tokens by content similarity and attends within each cluster, aggregating non-locally across the image. To

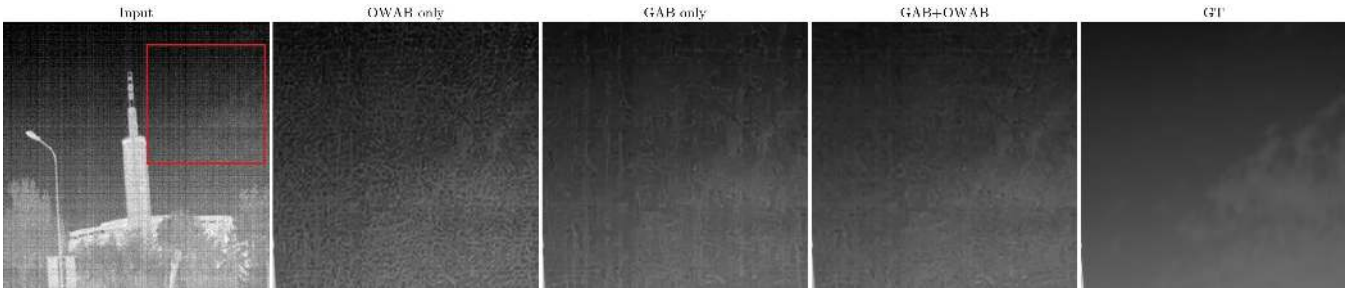


FIGURE 8. **Qualitative out-of-distribution result on a FLIR image. To emulate the stronger-than-training noise of real sensors, we degrade at $\sigma=10$, well beyond the Gaussian range $\sigma \in [0, 5]$ used during training. The first panel is the full degraded input, with the red box marking the crop region; the remaining panels are the square crop of each restoration, where the surrounding homogeneous sky reveals the residual noise. The window-only OWAB leaves grain-like noise in the sky, while the clustering-based GAB and the full model suppress it, matching the SSIM gap in Table 4.**

TABLE 3. **Ablation of the two Stage-1 attention modules (GAB, OWAB), reported as PSNR↑/SSIM↑ on the synthetic validation sets (100 FLIR and 20 M^{3}FD images) at the final training iteration. Table 2 instead evaluates the selected checkpoint on the full test sets, so the two tables are not directly comparable. Best.**

Module			FLIR		M^{3}FD	
GAB	OWAB	#Param	PSNR↑	SSIM↑	PSNR↑	SSIM↑
	✓	3.1M	35.55	0.948	36.93	0.960
✓		3.7M	35.28	0.947	36.51	0.961
✓	✓	4.7M	**35.81**	**0.949**	**37.37**	**0.961**

TABLE 4. **Out-of-distribution robustness on the synthetic FLIR set. To emulate the out-of-distribution noise that real sensors produce, we apply test degradation stronger than that seen in training (Gaussian $\sigma \in \{10, 15, 20, 25\}/255$ and stripe $\sigma \in \{5, 7.5, 10, 12.5\}/255$, versus Gaussian $\sigma \in [0, 5]/255$ at training), and report PSNR↑/SSIM↑ against the clean ground truth. Best, second best.**

GAB	OWAB	σ=10	σ=15	σ=20	σ=25
	✓	29.59/0.756	24.23/0.462	21.39/0.330	19.36/0.251
✓		29.45/0.853	24.19/0.528	21.49/0.353	19.65/0.270
✓	✓	**31.80/0.880**	**26.36/0.620**	**22.75/0.418**	**20.38/0.305**

assess their individual roles, Table 3 reports full-reference fidelity on the synthetic validation sets for three configurations: OWAB only, GAB only, and the full model (GAB+OWAB). The full model is the most faithful on both sets, showing that the two forms of aggregation are complementary. Under this in-distribution degradation the two single modules are close, with the local OWAB marginally ahead in PSNR and comparable in SSIM. The benefit of the GAB's content-based clustering does not manifest under this mild in-distribution noise; as analyzed next, it emerges under stronger, out-of-distribution noise, where averaging over content-similar tokens is what removes the sensor noise.

Robustness to out-of-distribution noise. Table 3 measures fidelity under the same degradation strength seen in training, where local detail dominates and the window-based OWAB alone is therefore slightly ahead of the clustering-based GAB. Real sensors, however, often produce noise far stronger than that seen during training. To probe this, we keep the blur fixed and push the additive noise well past the training range (Gaussian $\sigma \in \{10, 15, 20, 25\}/255$ together with stripe noise $\sigma \in \{5, 7.5, 10, 12.5\}/255$, versus Gaussian $\sigma \in [0, 5]/255$ during training), and measure full-reference fidelity against the clean ground truth (Table 4). The ranking reverses. The full model is now the most robust at every level, and its lead widens as the noise grows, while between the two single modules the clustering-based GAB overtakes the window-based OWAB. Even at the mildest level ($\sigma = 10$) the GAB preserves structure far better (SSIM 0.853 vs. 0.756), and the OWAB's marginal 0.14 dB PSNR advantage disappears and reverses as the noise strengthens. This shows that the clustering mechanism keeps the model stable under degradations stronger than those seen in training, which corresponds to the operating conditions of real infrared sensors. Fig. 8 confirms this visually at $\sigma = 10$. In the homogeneous sky around the structure, the window-only OWAB leaves grain-like residual noise, whereas the clustering-based GAB and the full model suppress it.

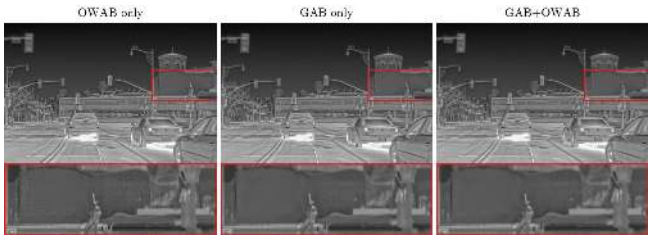


FIGURE 9. **Ablation on a real image: each panel is a full restoration with a zoom on a flat region. The full model (with the clustering-based GAB) suppresses the sensor noise more thoroughly than the variants without it.**

Clustering and noise removal. The GAB removes sensor noise by clustering similar tokens and averaging within each group; this non-local averaging is what cancels the independent noise. Fig. 9 compares the three variants on a real image, with a zoom on a flat region where the residual noise is most visible. The full model, which couples the clustering-based GAB with the OWAB, leaves the cleanest output, removing the sensor noise that the variant without clustering leaves behind. Since all three variants share a similar, compact parameter budget (3.1–4.7M), this gain comes from the clustering mechanism rather than from added capacity.

Effect of the diffusion stage. Table 5 isolates the contribution and the cost of Stage-2. By re-synthesizing fine textures that a

TABLE 5. Effect of the diffusion stage: quality–cost trade-off on the synthetic sets (full-image evaluation). FLOPs, time, and peak memory are measured for a 256 × 256 LR input (×4, 50 sampling steps), with time and memory on an RTX 3090.

	FLIR		M^3FD				
	PSNR↑	LPIPS↓	PSNR↑	LPIPS↓	FLOPs	Time	Mem
Stage-1 only	**35.96**	0.134	**37.29**	0.169	**1.0 T**	**0.28 s**	**3.0 GiB**
+ Stage-2	32.36	**0.113**	34.36	**0.137**	193.7 T	29.0 s	10.3 GiB

regression stage cannot restore, the diffusion stage improves LPIPS from 0.134 to 0.113 on FLIR and from 0.169 to 0.137 on M^3FD. These perceptual gains come at the expense of the distortion metrics, as PSNR drops from 35.96 to 32.36 dB, which is the familiar perception–distortion trade-off. The cost is dominated by the 50-step sampling, which raises the compute from 1.0 to 193.7 TFLOPs. Stage-1 alone produces a 1024 × 1024 output in 0.28 s within 3.0 GiB, about 3.6 outputs per second, comparable to the single-stage backbones of Table 2, whereas the full pipeline takes 29.0 s and 10.3 GiB on an RTX 3090. The two stages thus offer a practical choice between speed and quality, where Stage-1 provides fast and faithful restoration and Stage-2 is enabled when perceptual quality outweighs latency. Qualitative failure cases of the diffusion stage are analyzed in Fig. 12.

VI. DISCUSSION AND ANALYSIS

A. VALIDATION OF THE DEGRADATION MODEL

This subsection verifies the realism of the modeled noise through statistical analysis. Assessing a simulated noise model requires both a qualitative comparison against real infrared noise and a comprehensive statistical examination of its components. The real noise was acquired with a FLIR Boson 320 uncooled microbolometer. After the camera had run for more than 30 minutes so that the lens and housing were warm, a burst of consecutive RAW frames was captured with the lens cap closed. A representative frame is shown in Fig. 10(a). The infrared noise image produced by our physics-based degradation model is shown in Fig. 10(b), and the simulated and real noise exhibit visually similar spatial patterns. The separation and analysis of the individual noise components follow the protocol of Cai *et al.* [16], with lens-cap frames captured after prolonged operation used in place of blackbody imagery and the low-frequency hill bias removed before the stripe components are extracted.

We first extract the spatial noise by averaging the noise sequence in the temporal domain. Components that vary randomly over time cancel out during this averaging, so only the fixed-pattern noise remains in the temporally averaged image. Because stripe noise is a constant offset along each row or column, taking the per-row and per-column means of the averaged image after this hill removal yields the distributions of the row and column stripe noise. Their empirical distributions are compared with a normal distribution in the normal probability plots of Figs. 10(c) and (d). Evaluated with the Shapiro–Wilk and Jarque–Bera tests, both components give p-values above 0.05 (reported in Fig. 10), and thus the null

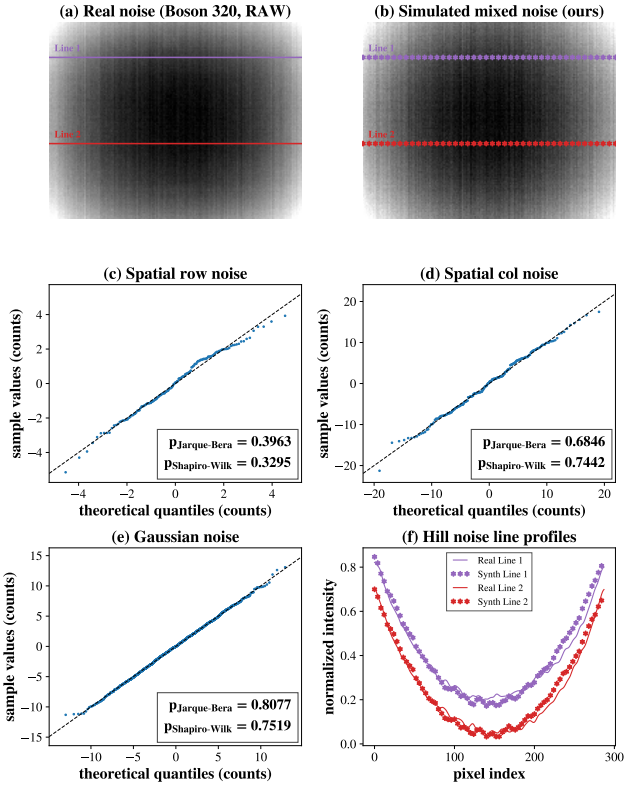


FIGURE 10. Noise-authenticity analysis of the composite degradation model. (a) Real noise captured by a FLIR Boson with the lens cap closed. (b) Infrared mixed noise simulated by our composite degradation model. (c) Normal probability plot of the spatial row noise. (d) Normal probability plot of the spatial column noise. (e) Normal probability plot of the Gaussian noise. (f) Pixel-value comparison of the hill noise along Line 1 and Line 2.

hypothesis of normally distributed noise cannot be rejected.

The Gaussian noise is then isolated from the difference of two adjacent frames. The fixed-pattern components shared by the two frames cancel in the difference, leaving only the temporally random Gaussian noise. Its normal probability plot is shown in Fig. 10(e), and both tests again fail to reject normality.

Hill noise originates from the uneven dissipation of heat across the sensor and the optics during operation and intensifies as the camera stays on. As visible in Fig. 10(a), the real hill noise is dark at the center and grows brighter towards the periphery, in agreement with the simulation. Fig. 10(f) plots the pixel values along Line 1 and Line 2 marked in (a) and (b), with solid lines for the real noise and asterisks for the simulated one. The real hill shows a slight asymmetry, likely due to imperfect mechanical and thermal symmetry of the physical sensor, yet the two distributions agree closely. Overall, our composite degradation model reproduces the noise characteristics of a real infrared sensor, indicating that models trained on the synthetic degradation can transfer to real-world infrared imagery.

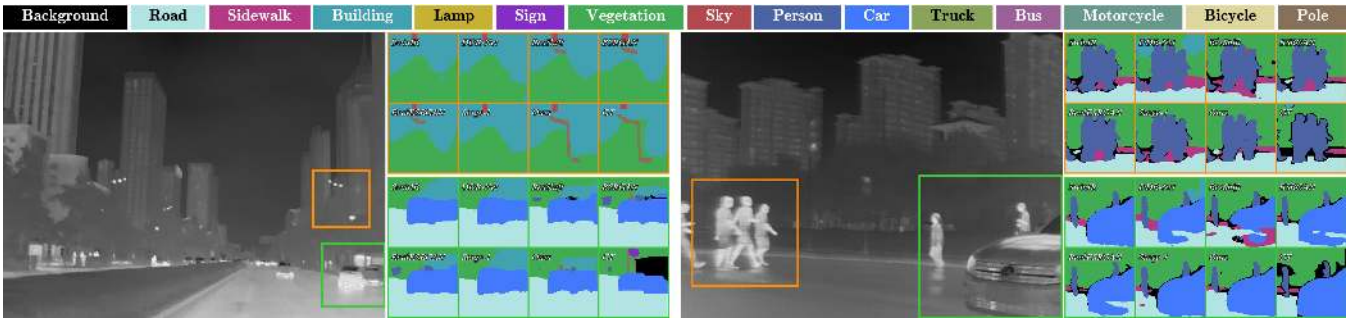


FIGURE 11. Qualitative segmentation comparison of infrared image super-resolution methods on the M^{3}FD dataset.

TABLE 6. Semantic segmentation results on the M^{3}FD dataset (mIoU ↑). Best and second best among SR methods.

Method	mIoU ↑
Bicubic	0.331
RRDBNet [54]	0.401
SwinIR [11]	0.398
DAT [31]	0.396
HAT [30]	0.393
CATANet [33]	0.396
IRLF-FSRNet [25]	0.394
DDR-Net [26]	0.379
BSRGAN [13]	0.416
Real-ESRGAN [14]	0.414
ResShift [35]	0.384
Ours (Stage-1)	0.399
Ours	**0.417**

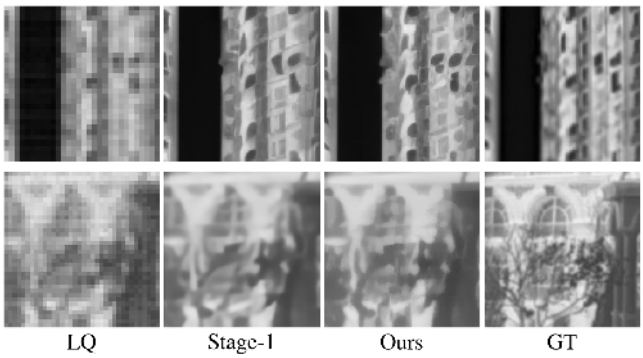


FIGURE 12. Failure cases of the diffusion stage.

B. DOWNSTREAM TASK EVALUATION

Setup. We perform semantic segmentation on the M^{3}FD dataset [55], using 280 infrared test images with 15 annotated classes. As the segmentation model, we use the FMB-pretrained network of SegMiF [66] and keep it frozen for all methods. Each image is degraded by the pipeline of Section III and downsampled $\times 4$. All methods restore the same degraded inputs using the checkpoints from Table 1, without any fine-tuning for the segmentation task, and mIoU is measured against the ground-truth annotations.

Quantitative. Table 6 presents the segmentation results. Our full model achieves the best mIoU of 0.417, comparable to BSRGAN (0.416) and Real-ESRGAN (0.414) and higher than the remaining baselines. It also outperforms Stage-1 by 1.8 mIoU points, which indicates that the diffusion stage does not introduce severe hallucination and its synthesized details remain consistent with the scene.

Qualitative. Fig. 11 presents a qualitative comparison of segmentation results. The other methods often fail to segment complete objects. In the left scene, none of them recovers the street-lamp pole, which only our model restores. The recovered pole matches the location and shape of the ground truth, showing that the details generated by the diffusion stage correspond to structures actually present in the scene rather than hallucinated ones. In the right scene, our model captures the shapes of the pedestrians and the car most completely, producing predictions closest to the ground truth.

C. FAILURE CASES OF THE DIFFUSION STAGE

This section analyzes when the diffusion stage fails, in terms of how much information the degraded input retains. When the structural cues survive in the input, the synthesized detail remains consistent with the scene, as shown qualitatively in Figs. 5 and 6 and confirmed at scale by the downstream gain in Section VI-B. The failures arise in the opposite case, where the $\times 4$ downsampling and the composite noise destroy the structural cues themselves. In the first scene the window and balcony grid of the apartment facade collapses at low resolution, so the conditioning becomes ambiguous and the diffusion stage turns the window rows into irregular blobs whose layout no longer matches the ground truth. In the second scene, degraded at the strongest intensities within the training range, the decorative structures of the palace facade are smoothed into featureless blobs that remove detail present in the ground truth. In both scenes the Stage-1 output remains blurry but does not introduce structures that conflict with the scene. Such failures concentrate predictably in regions where the structural cues are lost, for example dense clutter, fine repetitive patterns, and structures of only a few pixels, and are the spatial manifestation of the fidelity trade-off quantified in Table 5. The two-stage design anticipates them. Applications that require precise geometry or radiometric consistency can use the Stage-1 output directly, and the diffusion stage remains an optional refinement for applications where perceptual quality is the priority.

VII. CONCLUSION

In this paper, we studied blind super-resolution of real-world infrared images, whose sensors introduce a composite degradation that couples optical blur with Gaussian, stripe, and non-uniformity noise and varies in both composition and strength with the operating conditions. We proposed NLAN-Diff, a two-stage framework that separates robust noise suppression from texture synthesis. We first built a physics-based composite degradation model that injects each corruption at its originating stage of the imaging chain, giving a realistic training distribution for blind IISR. The first stage, the Non-Local Aggregation Network (NLAN), clusters self-similar thermal regions from across the image and averages within each group, realizing a learnable non-local means that suppresses the composite, out-of-distribution sensor noise while preserving structure. The second stage then conditions a frozen generative diffusion prior on this clean restoration to synthesize realistic fine textures.

Extensive experiments on synthetic benchmarks and real infrared images show that NLAN-Diff achieves the best MUSIQ scores across all four evaluation regimes and the best MANIQA scores on the synthetic sets, and that, used as a restoration backbone, NLAN reaches top-tier full-reference fidelity. Our ablations further confirm that the clustering mechanism, rather than added capacity, is what keeps the model stable under noise stronger than that seen in training. A few limitations remain. We focus on $\times 4$ super-resolution, and the diffusion stage adds to the inference cost, so extending to other scale factors and improving efficiency are natural next directions. Moreover, the real-world test sets are modest in size, with 47 Real-Light and 29 Real-Heavy images, have no ground truth, and are evaluated only with no-reference metrics such as MUSIQ and MANIQA, which were developed on natural RGB images and may not fully reflect infrared perceptual quality, so constructing larger real paired infrared benchmarks would allow a more complete assessment. Finally, since the diffusion stage relies on a clean conditioning image, the final quality is bounded by how faithfully the first stage removes the degradation, so producing an even cleaner high-resolution restoration is an important direction for strengthening the pipeline. Infrared imaging also underpins safety-critical applications such as industrial condition monitoring and intelligent fault diagnosis, where restored images may feed downstream diagnostic decisions and the black-box nature of a generative diffusion prior calls for robust and interpretable restoration, in line with the growing emphasis on explainable AI in intelligent fault diagnosis [67]. Our two-stage design offers a practical entry point for such systems, since the regression-based Stage-1 output preserves measurement fidelity while the generative refinement remains optional. We believe that decoupling robust restoration from generative refinement, as in NLAN-Diff, offers a practical path toward reliable super-resolution under the diverse degradations of real infrared sensors.

CONFLICT OF INTEREST

The authors declare no conflict of interest.

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